

Kartik Maheshwari

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EXPERIENCE

Data Associate

April 2026

Yal.ai

Remote

- Performed data annotation, research and collection for **AI models**, creating high-quality datasets enhancing training & evaluation processes.
- Collaborated with multiple teams to maintain quality standards, & support **real-time communication systems**

Undergraduate Intern

Jun 2025 – Jul 2025

Malaviya National Institute of Technology Jaipur

Jaipur, Rajasthan

- Generated and transformed large-scale 5G datasets into image-based representations, reducing dataset size by 99.6%, significantly lowering compute requirements for ML-based channel prediction.
- Deployed a real-time **OAI-based 5G testbed** to simulate and analyze Layer-3 and cross-layer attacks

Summer Intern

May 2025 – July 2025

Indian Institute of Technology Patna

Remote

- Deployed a **Streamlit** AI system for **1000+** students, eliminating seating conflicts via constraint logic.
- Automated seating plans generation and **Excel** attendance sheets, reducing time from **hours to minutes**.

PROJECTS

UrbanPulse | Under Patent | *IoT, Edge Processing, Embedded Programming, Piezoelectric Sensors*

Demo [↗](#)

- Collaborated on the software pipeline for a **100%** self-sustaining **IoT** pavement prototype, integrating edge-level signal processing with **cloud telemetry** for real-time traffic analytics.
- Developed edge-computing logic to process **real-time analytics** for smart **real-time congestion control**, contributing to reduction in reliance on traditional camera-based system's

Recruitment Management System | *AWS, Nginx, Cloudflare, Node.js, React*

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- Engineered a production-grade backend system with **Node.js** + **Redis** caching, reducing repeated DB/API hits by **40–60%** and integrated **Nginx** reverse proxy, server-side caching, and **AWS**-based deployment.
- Designed a fault-tolerant architecture with multiple failure-handling mechanisms, admin approval workflows, retry-safe APIs, validation layers, and **ExcelJS**-based data exports for reporting to non-IT stakeholders.

Netsense | *Python, Scapy, SQLite, scikit-learn*

Github

- Developed a **CLI-based packet analyzer** to capture and interpret real-time network traffic, implementing multi-layer threat detection via **deep packet inspection** for attacks such as **ARP** and **DNS spoofing**.
- Integrated an ML-based IDS pipeline leveraging the **CICIDS-2017** dataset for threat classification under **150ms**.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Kotlin, Java

Mobile: Jetpack Compose, Android SDK, React Native, Zustand

Full Stack: React, JavaScript, TypeScript, Tailwind, Node.js, Express, Redux, REST APIs, PostgreSQL MongoDB, MySQL, Redis

Tools: Git, GitHub, GCP, Firebase, Supabase, Postman, Docker, AWS, CI/CD, Vim, Android Studio, Figma, Linux

EDUCATION

Rajasthan Technical University

Kota, Rajasthan

B.Tech/B.E. in Computer Science and Engineering

CGPA: 8.8

ACHIEVEMENTS

Winner – Electrothon 8.0 (NIT Hamirpur) : Secured **1st Place** in a 48-hour national-level hackathon by **MLH**; developed a hybrid piezo and triboelectric-based intelligent IoT device. (**Under Patenting Process**)

Techgium (9th Edition) – Larsen & Toubro (L&T) : Qualified for Round 2 in L&T's national-level innovation competition, competing among top engineering students across India

Competitive Programming: Solved more than **250** DSA problems with a LeetCode rating of **1519**

Problem Solving and SQL Certificate: HackerRank | Certificate IDs - [452c7912f8e4](#), [fd5fd1d8235b](#)

Technical Head: Innovation and Incubation Cell (e-Cell)